higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

MARKING GUIDELINE

NATIONAL CERTIFICATE

MECHANICAL DRAWING AND DESIGN N5

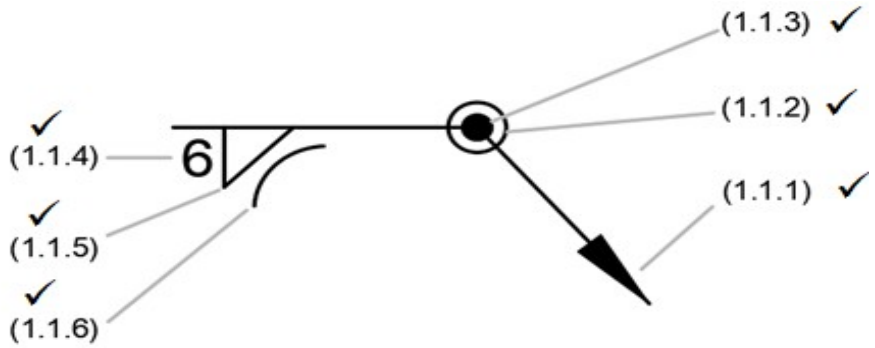
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QUESTION 1

1.1 STANDARD WELD SYMBOL



(6)

1.2 $tan\Theta = \frac{10}{20}$

$tan\Theta = 0,5$

$\Theta = 26,6^{o}$ ✓✓

$\sin 26,6^{o} = \frac{throat}{20}$

$Throat = \sin 26,6^{o} \times 20$

$Throat = 8,95 \ \ mm$

The throat can be said to be 9 mm.✓✓ (4)

[10]

QUESTION 2

2.1 Diameter of rivets

$d = 6\sqrt{t}$

$= 6 \times \sqrt{12}$

$= 6 \times 3{,}464 \ \ mm$

$= 20{,}784 \ \ mm$

Use 20 mm standard size rivets.✓✓ (2)

Ultimate stresses:

$\sigma_t = 480 \ \ Mpa$

$\tau = 384 \ \ Mpa$

$\sigma_c = 618 \ \ Mpa$

Working stresses:

$$Allowable \ \ stress = \frac{Ultimate \ \ stress}{Factor \ \ of \ \ safety}$$

$$\sigma_t = \frac{480 \times 10^6 N/m^2}{6} 80 \ \ MPa$$ ✓

$$\tau = \frac{384 \times 10^6 N/m^2}{6} = 64 \ \ MPa$$ ✓

$$\sigma_c = \frac{618 \times 10^6 N/m^2}{6} = 103 \ \ MPa$$ ✓ (3)

Pitch

Consider one pitch length of the joint.

For the best pitch:

Strength of plate in tension = strength of rivets in shear

$$\sigma_t \times (p - d) \times t \times \frac{\pi d^2}{4}$$

Use either ultimate or working stresses.

$$480 \times 10^6 N/m^2 \times (p - 0{,}02 \ \ m) \times 0{,}012 \ \ m = 384 \times 10^6 \ \ N/m^2 \times \frac{\pi \times (0{,}02 \ \ m)^2}{4}$$

$$p - 0{,}02 \ \ m = \frac{120 \ \ 637{,}2 \ \ N}{480 \times 10^6 \ \ N/m^2 \times 0{,}012}$$

$p = 0{,}012 \ \ m + 0{,}02 \ \ m$

$= 0{,}041 \ \ m$

$Pitch = 41 \ \ mm$ ✓✓

Check that this is above the allowable minimum

$Minimum \ \ pitch = 2 \times d$

$= 2 \times 20 \ \ mm$

$= 40 \ \ mm$ ✓✓ (4)

2.2 Failing loads per pitch length
Ultimate stresses must be used.

Plate in tension
Ultimate tensile load = Ultimate tensile stress x cross-sectional area of plate at rivet hole

$$P = \sigma_t \times (p - d) \times t$$
$$= 480 \times 10^6 \ N/m^2 \times (0{,}041 \ m - 0{,}02 \ m) \times 0{,}012 \ m$$
$$= 120{,}96 \ kN \checkmark\checkmark$$ (2)

Rivet in shear
Ultimate shear load = Ultimate shear stress x cross-sectional area of rivet

$$P = \tau \times \frac{\pi d^2}{4}$$
$$= 384 \times 10^6 \ N/m^2 \times \frac{(0{,}02 \ m)^2}{4}$$
$$= 120{,}64 \ kN \checkmark\checkmark$$ (2)

Crushing of rivets against the plate
Ultimate crushing load = Ultimate crushing stress x projected area of contact

$$P = \sigma_c \times d \times t$$
$$= 618 \times 10^6 \ N/m^2 \times 0{,}02 \times 0{,}012$$
$$= 148{,}32 \ kN \checkmark\checkmark$$ (2)

2.3 Failing load is therefore 120,6 kN and the joint would fail by shearing at the rivets.✓ (1)
[16]

QUESTION 3

$$Allowable\ \ tensile\ \ stress \in the\ \ belt = \frac{Allowable\ \ force \in the\ \ belt}{Cross\text{-}sectional\ \ area\ \ of\ \ the\ \ belt}$$

$$Allowable\ \ force \in the\ \ belt = Allowable\ \ tensile\ \ strength \in the\ \ belt$$

$= 1{,}38 \times 10^6\ \ N/m^2 \times 0{,}1\ \ m \times 0{,}01\ \ m$
$= 1\ \ 380\ \ N$✓
This is the pull on the tight side T_1✓ (2)

`$T_1 = 2{,}2\ \ T_2$

$$\therefore T_2 = \frac{T_1}{2{,}2}$$

¿ $\frac{1\ \ 380}{2{,}2}$

$= 6{,}273\ \ N$✓✓ (2)

$$Power\ \ transmitted = \frac{(T_1 - T_2)\ \times \pi\ (D\ +\ t)N}{60}$$

¿ $\frac{(1380\text{-}627{,}3) \times \pi (0{,}25 + 0{,}01) 800}{60}$

$= 8197{,}6\ \ watts$
$= 8{,}1976\ \ W$✓✓ (2)
[6]

QUESTION 4

4.1 DATA:

$F = 90\ \ kN;\ \ d = d_1 = 40\ \ mm;\ \ d_2 = 80\ \ mm;\ \ t = 30\ \ mm;\ \ t_1 = 15\ \ mm;\ \ SF = 5$

4.1.1 For ultimate tensile stress, consider rod:

$$F = \frac{1}{4}\pi d_1^2 \sigma_t$$

$$\therefore \sigma_t = \frac{4F}{\pi d_1^2} = \frac{4(90 \times 10^3)}{\pi (0{,}04)^2} = \ \ 71{,}6197 \times 5 = 358{,}09\ \ MPa$$✓

Consider eye:

$$F = (d_2 - d_1) t \sigma_t$$

$$\therefore \sigma_t = \frac{F}{(d_2 - d_1)t} = \frac{(90 \times 10^3)}{(0{,}08\text{-}0{,}04)(0{,}03)} = 75{,}000 \times 5 = 375{,}000\ \ MPa$$✓ (4)

Ultimate tensile stress = 375 MPa✓✓

4.1.2 For ultimate shear stress, consider pin:

$$F = 2\left(\frac{1}{4}\pi d^2\right)\tau$$

$$\therefore \tau = \frac{4F}{2\pi d_1^2} = \frac{4\left(90\times10^3\right)}{2\pi\left(0{,}04\right)^2} = 35{,}81\times5 = 179{,}05 \;\; MPa\checkmark\checkmark$$

(2)

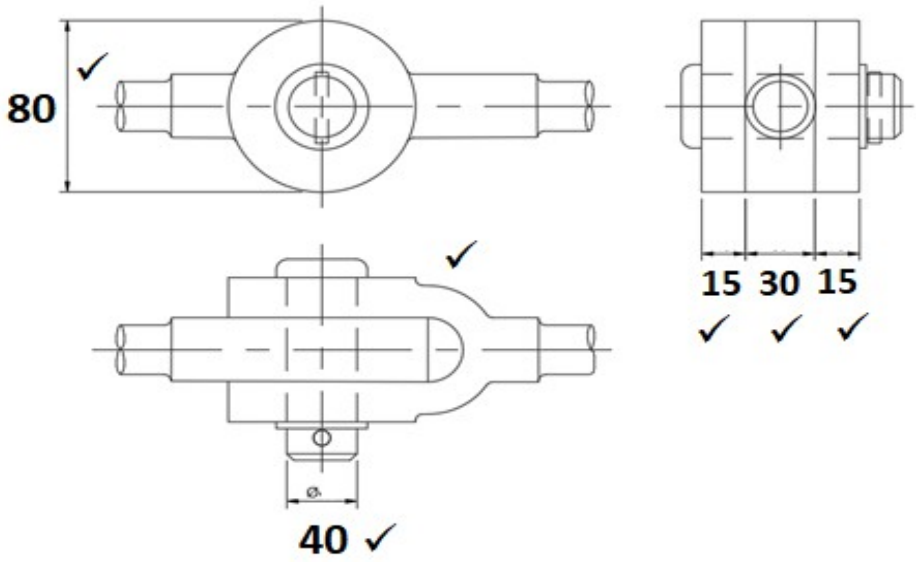
4.1.3 For ultimate compressive stress, consider eye:

$$F = d_1 t\sigma_c$$

$$\therefore \sigma_c = \frac{F}{d_1 t} = \frac{\left(90\times10^3\right)}{\left(0{,}04\right)\left(0{,}03\right)} = 75{,}00\times5 = 375 \;\; MPa\checkmark\checkmark$$

(2)

4.2 THREE VIES OF KNUCKLE JOINT



(6)

4.3 PURPOSE
Used to join two parts or rods, where one may have a small angular or axial movement about the other.

APPLICATIONS
- Machinery: valve mechanisms, reciprocating engines✓
- Structures: suspension chains, bridges, roof trusses, braced girders✓
- Joints of gearing✓ and elevator chains✓ (4)

[18]

QUESTION 5

5.1 Maximum load on piston rod = Area of piston × steam pressure

$$P = \frac{\pi D^2}{4} \times p$$

$$= \pi \times \frac{(0{,}762\,m)^2}{4} \times 1{,}03 \times 10^6\,N/m^2$$

$$= 469{,}72\ \ kN$$ ✓✓ (2)

For the rod load = Area of rod x stress in rod

$$P = \frac{\pi d^2}{4} \times \sigma$$

$$d = \sqrt{\frac{P \times 4}{\pi \times \sigma}}$$

$$¿ \sqrt{\frac{469{,}72 \times 10^3\,N \times 4}{\pi \times 27{,}5 \times 10^6}}$$

$$d = 0{,}1476\ \ m$$

Say d = 148 mm✓✓ (2)

5.2 A taper of 1 in 4 on diameter means the diameter decreases 1 mm for each 4 mm in length.
Therefore, in 152 mm length the diameter decreases:

$$1 \times \frac{152}{4} = 38\ \ mm$$

Diameter at end of taper = $152 - 38 = 114\ \ mm$✓✓ (2)

Threaded portion

$$Nominal\ \ diameter\ \ of\ \ bolt = Core\ \ diameter + 1{,}227 \times pitch$$
$$Core\ \ diameter = Nominal\ \ diameter - 1{,}227 \times pitch$$

Assume a pitch of 6 mm

$$d_r = 102 - 1{,}227 \times 6$$
$$= 94{,}638\ \ mm$$

Say 95 mm✓✓ (2)

$$Tensile\ \ stress \in threaded\ \ portion = \frac{load}{area}$$

$$\sigma_t = \frac{P}{\frac{\pi d_r^{\,2}}{4}}$$

(2)

$$\sigma_t = \frac{469{,}72 \times 10^3\ N \times 4}{\pi \times (0{,}095\ m)^6}$$
$$= 66{,}27\ \ MPa \checkmark\checkmark$$

[10]

QUESTION 6

6.1 The diameter of the shaft

$$Power = \frac{2\pi NT}{60}$$

$$\therefore T = \frac{Power\ \times\ 60}{2 \times \pi \times N}$$

$$¿ \frac{80 \times 10^3 \times 60}{2 \times \pi \times 290\ \ r/min} \checkmark$$

$$= 2\ \ 634\ \ Nm \checkmark\checkmark$$ (3)

$$T = \frac{\pi d_s^{\ 3}}{16} \times \tau$$

$$\therefore d_s^{\ 3} = \frac{T \times 16}{\pi \times \tau}$$

$$¿ \frac{2631\ \ Nm \times 16}{\pi \times 62 \times 10^6\ N/m^2}$$

$$¿ \sqrt[3]{0{,}0002162\ \ m^2}$$

$$= 0{,}06\ \ m \checkmark$$

$$= 60\ \ mm \checkmark\checkmark$$ (3)

6.2 The length of the claws

$$D^2 - d^2 = \frac{\pi d_s^{\ 3} \times \tau_{shaft}}{2 \times n \times l \times \sigma_{cclaws}}$$

$$l = \frac{\pi d_s^{\ 3} \times \tau_{shaft}}{2 \times n \times \sigma_{cclaws} \times (D^2 - d^2)}$$

$$¿ \frac{\pi \times (0{,}06\ m)^3 \times 62 \times 10^6\ N/m^2}{2 \times 3 \times 12 \times 10^6\ N/m^2 \times [(0{,}16\ m)^2 - (0{,}08\ m)^2]} \checkmark$$

$$= 0{,}0304 \checkmark$$

Say length = 31 mm ✓✓ (4)

[10]

QUESTION 7

7.1 Thickness of cylinder walls = $\frac{P_i \times D}{20,6 \times 10^6 N/m^2} + 0,008\ m$

$\therefore \frac{1,24 \times 10^6 N/m^2 \times 0,45\ m}{20,6 \times 10^6 N/m^2} + 0,008\ m$

$= 0,035\ m$✓

$t = 35\ mm$✓✓ (3)

7.2 Thickness of the flanges $= 1,2\ t$ ¿ $1,5\ t$

$= 1,2 \times 35\ mm$ ¿ $1,5 \times 35\ mm$

$A = 42\ mm$ ¿ $52,5\ mm$

Say 45 mm✓✓ (2)

7.3 Thickness of cylinder cover $= t + 10\ mm$

$= 35\ mm + 10\ mm$

$= 42\ mm$✓✓ (2)

7.4 Allowable load per stud = Core area of stud x allowable tensile stress

$= 144 \times 10^{-6} m^2 \times \frac{464 \times 10^6}{8}$

$= 8\ 352\ newtons$✓

$8\ 352\ newtons = \frac{197,21 \times 10^3\ newtons}{N}$

$N = \frac{197,21 \times 10^3\ newtons}{8352\ newtons}$

$N = 26,6$

Say 27 studs✓

$PCD = D + 2t + d$

$= 450\ mm + (2 \times 35\ mm) + 16\ mm$

$= 536\ mm$✓

$Pitch = Circumference\ of\ pitch \div Number\ of\ studs$ ¿

¿ $\frac{\pi \times 536\ mm}{27}$

$= 62,37\ mm$✓

Pitch to be approximately

$4 \times d$

$= 4 \times 16 \ \ mm$
$= 64 \ \ mm$✓✓ (6)

7.5 Width

$= 3 \times d$

$= 3 \times 16 \ mm$

$= 48 \ mm$✓✓ (2)

[15]

QUESTION 8

8.1 $d_r = d_s - 2h$

$= 0,058 \ m - 0,0055 \ m$

$= 0,047 \ m$

$A = \frac{1}{2}(0,058 \ m - 0,047 \ m) \times 10 \times 0,045 \ m$

$= 2,475 \times 10^{-3} \ m^2$

$r_m = \frac{d_s + d_r}{4}$

$= \frac{0,058 \ m - 0,047 \ m}{4}$

$= 0,02625 \ m$✓

$T = p \times A \times r_m$

$= 5 \times 10^6 \ N/m^2 \times 2,475 \times 10^{-3} m^2 \times 0,0265 \ m$

$= 324,8 \ Nm$✓✓ (3)

$Power = \frac{2\pi TN}{60}$

$= \frac{2 \times \pi \times 324,8 \times 2 \ 500}{60}$

$Power = 85,03 \ kW$✓✓ (2)

8.2 $Force \ on \ splines = p \times A$

$= 5 \times 10^6 \ N/m^2 \times 2,475 \times 10^{-3} m^2$

$= 12 \ 375 \ N$✓

$Sliding \ force = force \ on \ splines \times friction \ coefficient$

$= 12 \ 375 \times 0,1$

$= 1 \ 237,5 \ N$✓ (2)

[7]

QUESTION 9

$Friction\ \ force = \mu \times Q$
$= 0{,}06 \times 60 \times 10^3$ ✓✓

$Distance\ \ moved = \pi \times d \times N$
$= \pi \times 0{,}15 \times 60$ ✓✓

$Work\ \ absorbed = \pi \times Q \times \pi \times d \times N$
$= 0{,}06 \times 60 \times 10^3\ N \times \pi \times 0{,}15\ \ m \times 60\ \ r/min$
$= 101{,}8 \times 10^3\ Nm/min$ ✓✓

$Power\ \ absorbed\ \ by\ \ friction = \frac{101\ \ 800}{60}$

$= 1{,}7 \times 10^3\ \ W$
$= 1{,}7\ \ kW$ ✓✓

(4 × 2) **[8]**

TOTAL: 100

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